



CANDIDATE
NAME

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CLASS

2T

INDEX
NUMBER

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BIOLOGY

Paper 3

9744/03

12 SEPT 2022

2 hours

Candidates answer on the Question Paper.

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs. Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question paper.

Section B

Answer **one** question in this section on writing papers provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten your answers for **Section B** securely.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1 [25]	
2 [12]	
3 [13]	
4 or 5 [25]	
TOTAL	75

Section A

Answer **all** questions in this section.

Answer **all** questions.

- 1 Climate change influences how mosquito-borne diseases such as dengue, chikungunya and zika infection have spread across the world vectors such as mosquitoes. These mosquitoes generally have a narrow temperature tolerance and there is evidence to show the spread of such diseases migrating from tropical to sub-tropical regions due to global warming.

The effects of temperature on the rate of reaction are expressed as the temperature coefficient (Q_{10}). Changes in temperatures affect enzyme activity which will affect insect's growth and development.

- (a) Define what is Q_{10} theory and explain why insects have a narrow temperature tolerance range as compared to mammals.

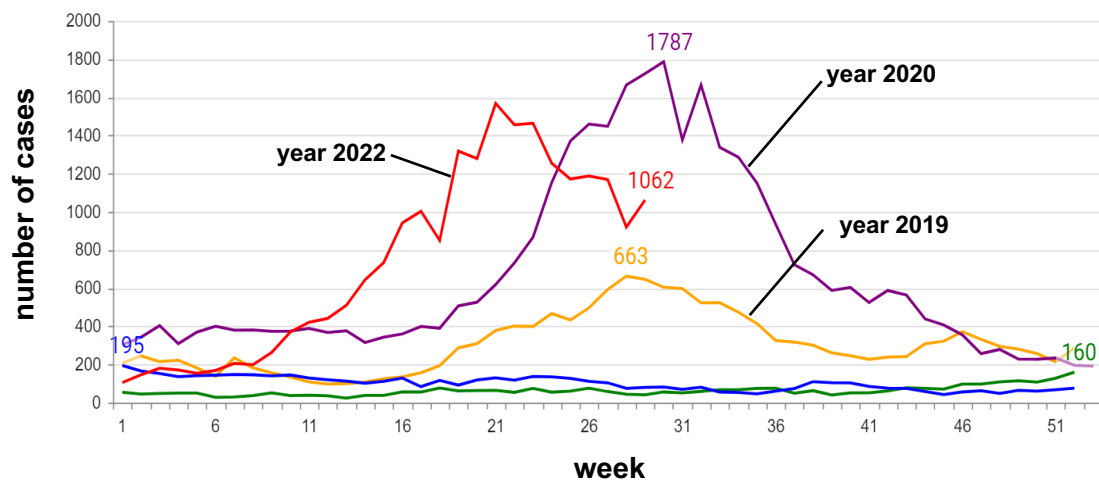
ANS [L1]

[2]

1. The Q_{10} theory of enzyme action states that for **every rise of 10°C** there is a **two-fold / twice increase** in enzyme action
2. Mosquitoes are **ectothermic / has a body temperature the same as the ambient temperatures** and any extreme temperatures beyond its physiological tolerance of that insect, may be lethal to the pathogen. / **OWTTE**

Singapore's dengue cases could rise even higher than the current number as the nation has entered the traditional peak dengue season, which usually lasts from June to October, corresponding to hotter and wetter climate.

Fig 1.1 shows the weekly number of dengue cases in Singapore from 2018 to 2022.



(Source: NEA website, July 22 - <https://www.nea.gov.sg/dengue-zika/dengue/dengue-cases>
Communicable Diseases Division, Ministry of Health)

Fig. 1.1

- (b) Explain how Singapore's hotter and wetter climate affects mosquitoes.

ANS [L2]

[3]

Any 3

1. Hotter and wetter climate in Singapore increases temperature which correspond to physiological changes of mosquito vector / OWTTE
 - a. Increased in development or growth of the mosquitoes / increase in development of females that spread dengue to humans / OWTTE
 - b. Shortening of breeding cycle of mosquitoes / shortening of mosquito's life cycle / OWTTE

- c. Prolonging bite rates and transmission seasons / increase in body size and wingspan which can travel further / OWTTE
- d. Possible evolution of a more heat tolerant mosquitoes thereby increasing pathogen range / OWTTE
- e. More breeding grounds for female mosquitoes to reproduce / lay eggs / OWTTE

Singapore has been looking on ways to reduce the numbers of *A. aegypti*. Besides project *Wolbachia*, Singapore may consider another type of genetically modified (GM) male mosquitoes. One of the genes added to these mosquitoes, when switched on, results in the production of a protein which is toxic to mosquitoes.

(c) GM male mosquitoes were released into the wild to mate with females and all the resulting offspring died in the larval stage.

(i) Suggest why releasing such large numbers of male mosquitoes did not immediately increase the risk of transmission of dengue fever.

ANS [L1

[1]

(Any 1)

(Any 1)

Male mosquitoes do not bite humans /

Only female mosquitoes bite prior to laying eggs

(ii) Both GM male and non-GM male mosquitoes were released to compare their dispersal and life span in the wild. The GM mosquitoes could be identified because they also carried a gene for green fluorescent protein (GFP).

Suggest why, in many examples of gene technology, fluorescent markers are used.

ANS [L2]

[1]

1. It is easier to **identify / screen** which ones are GM mosquitoes from the wild type ones, and this is **economical / convenient / time - saving / OWTTE**

In addition to the gene for GFP, the DNA that has been added to the GM mosquitoes consists of

- a promoter
- a gene coding for a toxic protein, tTA
- a binding site for tTA.

When a GM mosquito larva hatches from an egg, the promoter induces the production of only a small amount of tTA, so that the larva does not die immediately. In a process of positive feedback, the tTA produced binds to the DNA as shown in Fig. 1.4. This increases the expression of the gene until the increased concentration of tTA kills the larva.

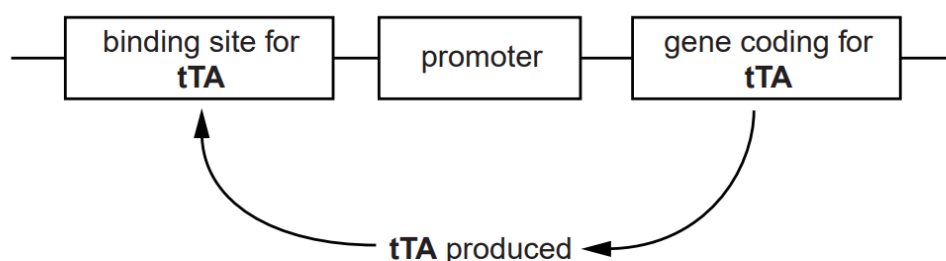


Fig. 1.4

- (d) Explain why this process is called positive feedback and why all 3 components shown in Fig. 1.4 are necessary for positive feedback to occur.

ANS [L2]

[3]

1. **Expression of tTA gene** to tTA protein causes **an increased expression of more tTA protein OWTTE**

AND (2 points below)
2. Promoter is needed to **initiate transcription / switches on gene / causes gene expression** where **RNA polymerase and (general) transcription factors will bind to**
3. The **binding site for tTA** near the promoter is a control element **necessary for upregulating the expression of tTA gene** upon binding of the tTA protein, which allows for positive feedback to occur.
4. The three components need to be introduced together, otherwise the gene may not be regulated by the tTA binding site / undergo normal expression without positive feedback

Dengue fever is a disease caused by infection with a virus transmitted by the *A. aegypti* mosquito. It is an acute viral infection characterised by fever, rash, headache, and muscle and joint pain. Occasionally, dengue virus infections progress to dengue haemorrhagic fever (DHF) and dengue shock syndrome (DSS). Fig 1.5 shows the structure of a dengue virus. Upon entry to host's cells, viral replication and translation begins.

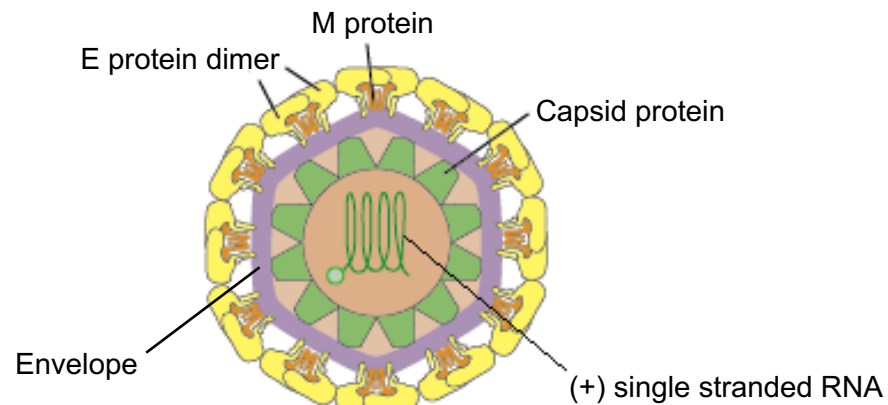


Fig. 1.5

(e) With reference to Fig. 1.5 and your knowledge of influenza virus,

- (i) state two ways in which the structure of dengue virus is similar to that of influenza virus.

ANS [L1]

[2]

1. Presence of viral envelope
2. Presence of glycoproteins on the viral envelope
3. Presence of capsid
4. Genome is single-stranded RNA
5. **AVP**

- (ii) outline how dengue virus replicates within the host's cells to form more viral genome and proteins.

ANS [L2]

[2]

1. Upon entry, positive single-stranded viral RNA attaches itself to **ribosomes** of RER and translated viral proteins / using **host cell machinery** to translate or synthesise viral proteins
2. Viral RNA is replicated to form negative single-stranded / negative sense / complementary strand RNA first in order to serve as template to create more copies of positive sense RNA viral genome

Dengue viruses consist of four serotypes, DENV-1 to DENV-4. The rapid identification of dengue virus serotypes isolated from patients' blood is important for clinical investigations. Individuals who become infected with one serotype obtain lifelong immunity against that serotype but not the other three.

Fig. 1.6 shows an IgM Antibody Capture Enzyme-Linked Immunosorbent Assay (MAC-ELISA), a molecular tool used to test on dengue patients, who developed anti-DENV antibodies. The tool includes:

- 1) the microtiter plate with anti-human-IgM antibody,
- 2) DENV antigens,
- 3) anti-DENV antibody conjugated with enzyme,
- 4) substrate which can change colour.

non-coloured
substrate

coloured
substrate

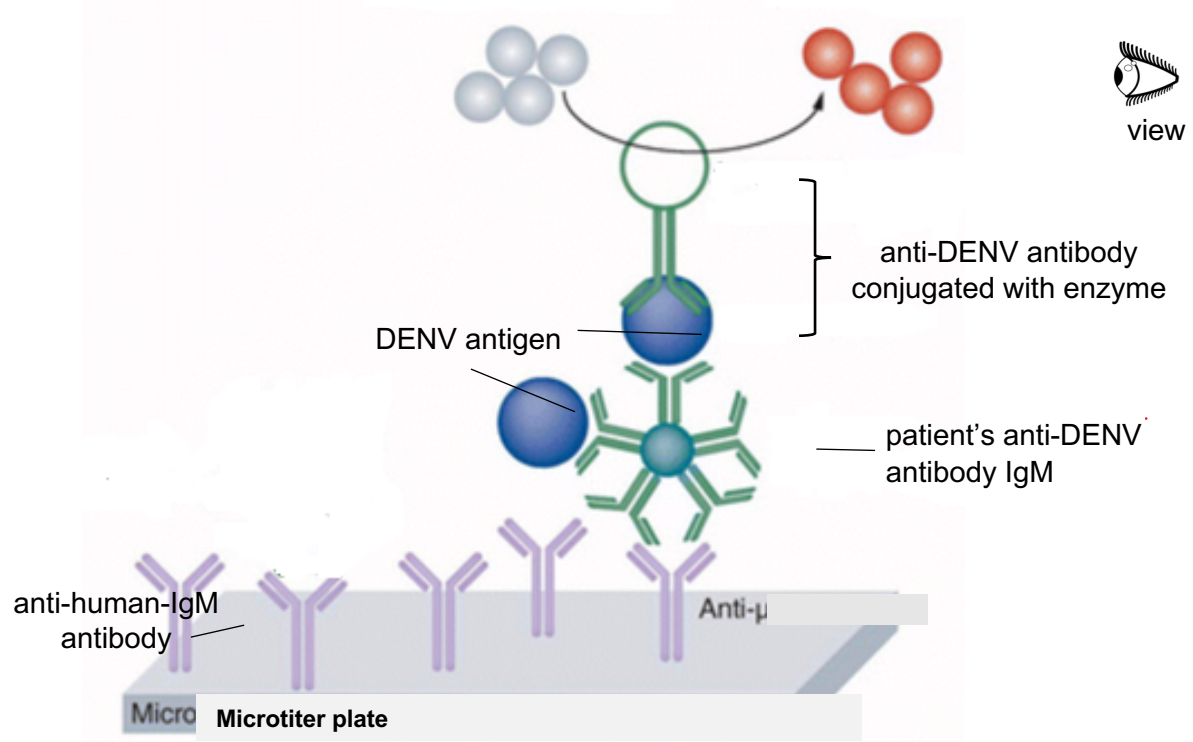


Fig. 1.6

- (f) With reference to Fig. 1.6, suggest the principles of using MAC-ELISA to detect a positive result of a patient infected by the dengue virus

ANS [L3]

[3]

1. MAC-ELISA is based on **complementary binding** of human IgM antibodies to that anti-human-IgM antibodies on a microtiter plate / to that DENV antigens within the tool,
2. Patients infected with DENV **will produce anti-DENV antibodies upon activation of B-cell into plasma cells**, which would bind to the DENV antigens and anti-human-IgM antibodies on the microtiter plate (or reverse argument)
3. **Enzymes conjugated to antibodies** would also bind to the DENV antigens of infected patients, which **can convert substrate to colour, making it easy to detect / to see / for positive result**

- (g) Dengue fever is considered an infectious disease, different from other non-infectious diseases like genetic diseases.

Comment on the differences between an infectious disease such as dengue fever to that of a genetic disease, using a named example.

ANS [L3]

[3]

	Features	Infectious disease	Genetic disease
1	definition	It is caused by viruses or bacteria, which are contagious in nature and can spread / transmit (idea of transmission) from person to person through various ways such as body fluids and vectors.	It is a genetic disorder where there are mutations within the gene sequences / chromosomal abnormalities within the genomic sequences that interferes with the normal functioning of desired protein
2	example	e.g. dengue fever R! other infectious disease cited	e.g. sickle cell anaemia / AVP

3	timespan	Mostly acute in nature where diseases develop suddenly and last a short time, often a few days or weeks	May be chronic in nature where diseases are acquired may be since born and last over extended period of time, often months or years
4	inheritance	Non-inherited in nature only	Inherited and non-inherited in nature
5		AVP; - Triggers innate/adaptive immunity	- Does not trigger immunity

Scientists have analysed the respiration rates on crickets, *Hophlosphyrum griseus*, another type of insect, subjected to three different ambient temperatures. Fig. 1.7 shows the metabolic rate of crickets was significantly correlated to their body mass.

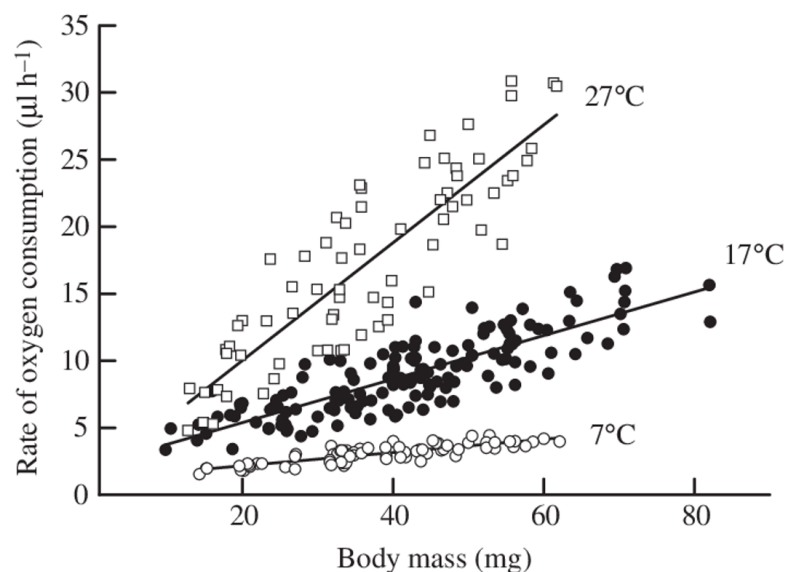


Fig. 1.7

(h) With reference to Fig. 1.7,

- (i) describe the extent to which a cricket's body mass can predict their rate of oxygen consumption at different ambient temperatures.

ANS [L3]

[3]

Any 3

- [Trend] From Fig. 1.7, the general trend shows that at body mass of 60 mg (choose a fixed body mass), as **temperatures increases** from 7°C to 17°C to 27°C, **rate of oxygen consumption increases** $4\mu\text{l h}^{-1}$ to $12\mu\text{l h}^{-1}$ to $27\mu\text{l h}^{-1}$ / OWTTE (depending on what body mass value quoted)
- It is observed that the **scattering of body mass data is wider** if the crickets are **subjected to higher temperatures** (17°C and 27°C), as opposed to at lower temperatures (7°C) /OWTTE
- [Magnitude] The **magnitude on the rate of oxygen consumption** for crickets at **higher temperatures** (17°C and 27°C) is **greater than** that of **lower temperatures** (7°C), which shows $11\mu\text{l h}^{-1}$ at 17°C and $22\mu\text{l h}^{-1}$ at 27°C, but only $2\mu\text{l h}^{-1}$ at 7°C.
- [Trend] An **increase in body mass**, shows an **increase in rate of oxygen consumption** in all 3 diff. temperatures, evident from the **positive gradient** in the graph /OWTTE
- AVP**

- (ii) Outline the role of oxygen in oxygen consumption of these crickets.

ANS [L2]

[2]

(Any 2)

1. Oxygen is the **final / last electron acceptor** at the end of ETC where it combines with electrons and protons to **form water**
2. It helps to **maintain flow of electrons along ETC** to **release energy for chemiosmosis / OWTTE**
3. and to **regenerate NAD⁺ and FAD** for glycolysis, link reaction and Krebs cycle

[Total: 25]

2 Stem cells have an important role in maintenance and growth within the bodies of eukaryotic organisms.

(a) Explain how stem cells maintain their presence in tissues.

ANS [L1]

[2]

Annotate:

1. Stem cells **divide by mitosis** producing 2 genetically identical daughter cells
2. Subsequently **one daughter cell immediately differentiates** while the **other is maintained as a stem cell**.
3. Self-renewal / maintenance of Stem cell line

(b) Normal differentiated cells usually have a limit to the number of cell divisions that can take place before apoptosis. Explain why stem cells are different and are able to divide indefinitely.

ANS [L2]

[2]

Annotate:

1. Stem cells possess **active telomerase** throughout, which **synthesize telomeres** at the 3' ends of parental DNA.
2. Telomeres act as a **buffer against the End replication problem** which sees the DNA shortening in each subsequent generation of daughter cells. In this manner Stem cells are able to divide continuously. OWTTE

Embryonic stem cells, which are derived from the inner cell mass of mammalian blastocysts, are used in research to understand disease development and develop future cell-based therapies for currently untreatable diseases.

In 2006, a team of scientists led by Shinya Yamanaka discovered that differentiated cells can be isolated and modified to an embryonic-like state. These cells, subsequently designated as induced pluripotent stem cells (iPSCs), have been hailed as an effective replacement for human embryonic stem cells for its usefulness in regenerative medicine. Fig. 2.1 shows a potential application of human iPSCs in gene therapy.

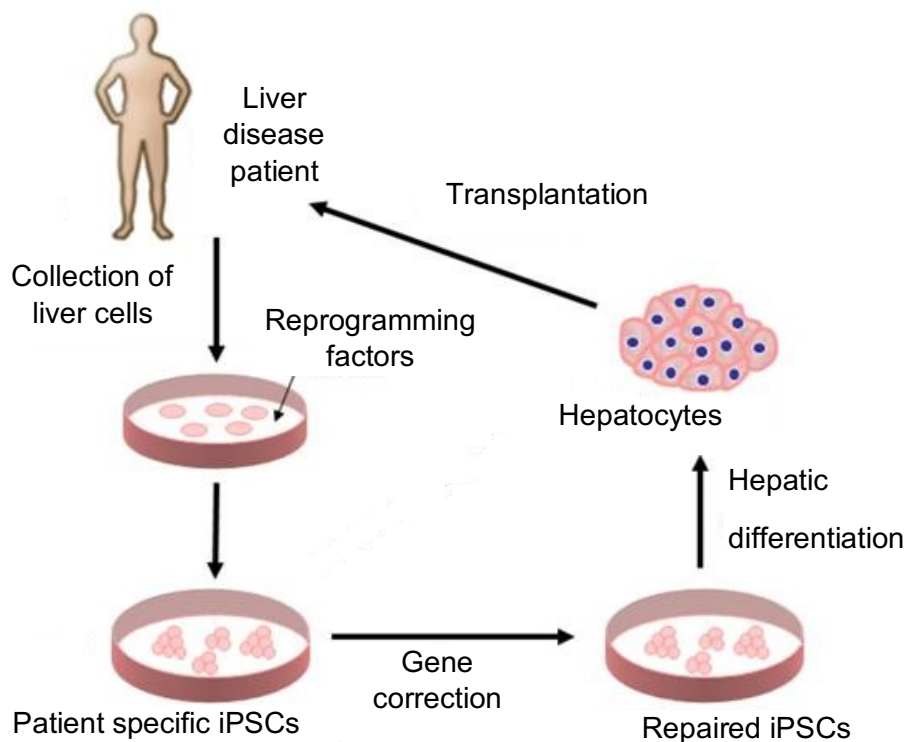


Fig. 2.1

- (c) Embryonic stem cells and iPSC are both pluripotent. In terms of genetic control, suggest how this is possible.

ANS [L3]

[2]

Annotate:

1. **Methylation** band patterns are **similar** OWTTE
2. Such that basic house keeping genes / that run the normal functions of the cell are kept "on" while other genes that determine specialisation are silenced. OWTTE

- (d) With reference to Fig. 2.1 state and explain two advantages of using iPSCs in gene therapy.

ANS [L1]

[2]

Annotate:

1. iPSCs can be derived from **differentiated cells taken directly from the patient** (and will therefore be **genetically identical** to the patient);
Hence there is **reduced immune incompatibility** between donor cells and recipient / lower chance of rejection of transplanted cells;

R! Tissue rejection

2. iPSCs can be **obtained in great numbers** from **any differentiated somatic cell**, allowing for a greater ease in which **positive selection of repaired iPSCs** can take place.
3. iPSCs are **not derived from human embryos** / does not involve isolation and subsequent death of embryos;
Hence **avoiding ethical issues associated with usage of embryos** in scientific research (and scientists are more likely to obtain federal funding and support) OWTTE
4. It also has fewer ethical issues involved, as **genetic makeup of patient's descendants are not altered**, unlike in **germ-line gene therapy**

Max 2

- (e) Yamanaka's research team studied 24 genes expressed by embryonic stem cells in mice to track down genes that can induce pluripotency. They discovered that four genes, notably *Oct4*, *Sox2*, *Klf4* and *c-Myc*, encode transcription factors which could be used to reprogram differentiated cells to form iPSCs. These genes can be introduced into differentiated somatic cells via viral transduction or the introduction of non-integrating plasmids in Fig. 2.2.

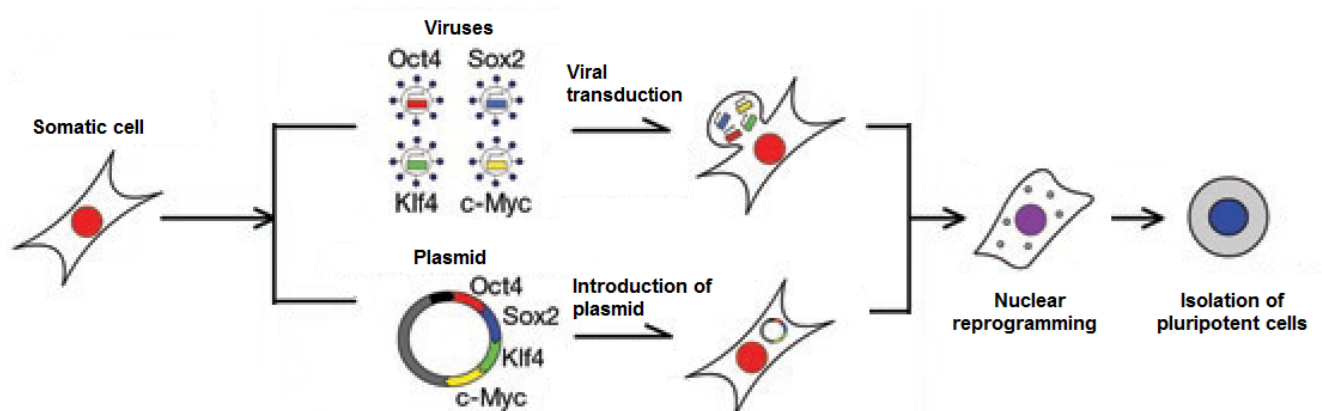


Fig 2.2

- (i) Suggest why it is necessary to introduce these four genes into somatic cells in iPSC formation.

ANS [L3]

[2]

Annotate: why necessary 4 genes iPSC formation
1] Diff cells 4 genes silenced
2] Code for transcription factors – silence genes.

1. These four genes are **not expressed** / The four genes are **methyated in differentiated** (somatic) cells
2. The four genes code **for transcription factors** which trigger the **silencing** of specific genes
OWTTE
R! Genes are not found in somatic cells

- (ii) Describe one advantage and one disadvantage of introducing the four genes via non-integrating plasmids over viral transduction using retroviruses.

ANS [L2]

[2]

Annotate: why necessary 4 genes iPSC formation

1. Advantage #1: **Plasmids do not integrate** into the genome, hence it does not cause insertional mutagenesis which can facilitate cancer development
2. Advantage #2: **Lower risk of stimulating immune response**
3. Advantage #3: **Do not cause disease** / lower risk of pathogenicity
Max 1
4. Disadvantage #1: Plasmids do not integrate into the genome, hence **expression of transgenes is short-lived** / cells may only **achieve pluripotency temporarily** / require **multiple rounds of introduction** of plasmid
5. Disadvantage #2: **Lower efficiency of transferring transgenes** into cells
Max 1

[Total: 12]

- 3 Viruses are often considered to be in the grey area of life. Over the last 15 years or so, giant viruses found in amoebae have complicated their status as simple non-living structures. Mimiviruses and megaviruses can contain more genes than a simple bacterium and may encode genes for information storage and processing. Genes common to the domains Archaea, Bacteria and Eukarya can be found in different giant viruses, and some researchers argue on this basis that they constitute a fourth domain of life.

This view challenges the cell theory, where the three tenets are as described below:

1. All living organisms comprise one or more cells.
2. The cell is the smallest unit of life.
3. All cells arise from pre-existing cells.

(a) Discuss how viruses challenge the cell theory and concepts of what is considered living.

[L1/2]

[4m]

A] Argue that viruses could be considered living:

1. **Viruses could be considered living because they are made of biomolecules** and include genes that encode for proteins and even processing of genetic information.
2. Their **genome may contain genes common to organisms that are living** i.e. the current three domains of life.
3. They are able to **replicate to produce more copies of themselves**, although they require a host cell to do so.

B] Hence, viruses challenge the cell theory because if they are considered to be living organisms:

1. then **living organisms could be made of just proteins and nucleic acids** and not cells
2. **they would be the smallest unit of life** instead of cells as most viruses are much smaller than cells.
3. Viruses (as the simplest life form) **requires another life form to replicate itself**, and cannot replicate by itself;

C] Moreover, they would challenge the following concepts of what is considered living:

	Concepts of living	
1	Homeostasis / Excretion	Viruses do not perform homeostasis or excretion as they have no metabolic wastes or toxins and do not need to remove these from their main bodies to maintain a constant internal environment.
2	Movement	Viruses cannot move by themselves, they move solely by diffusion and are carried by vectors such as air, body fluids or water.
3	Nutrition / Metabolism	Viruses are able to exist in a metabolically inert extracellular state for extended periods of time , but living organisms cannot live without respiration / food for such long periods.
4	Growth	Viruses do not grow more complex after assembly; they only comprise a genome and proteins which do not change unless there is mutation in the genome.
5	Sensitivity	They are unable to sense the environment as they have no sense structures , such as chemical/ pressure receptors, and therefore do not respond to a change in environment until they come into contact with suitable host cells.

- (b) Despite being considered as non-living things, viruses are known to mutate and evolve into different strains. Concerns over the effectiveness of vaccines against the inevitable evolution of the coronavirus broke out amidst the pandemic.

In Singapore, one of common type of vaccine used for the coronavirus is also known as the mRNA vaccine.

1. First, mRNA COVID-19 vaccines are given in the upper arm muscle or upper thigh, depending on the age of who is getting vaccinated.
2. After vaccination, the mRNA will enter the muscle cells.
3. Once inside, they use the cells' machinery to produce a harmless piece of what is called the spike protein. The spike protein is found on the surface of the virus that causes COVID-19.

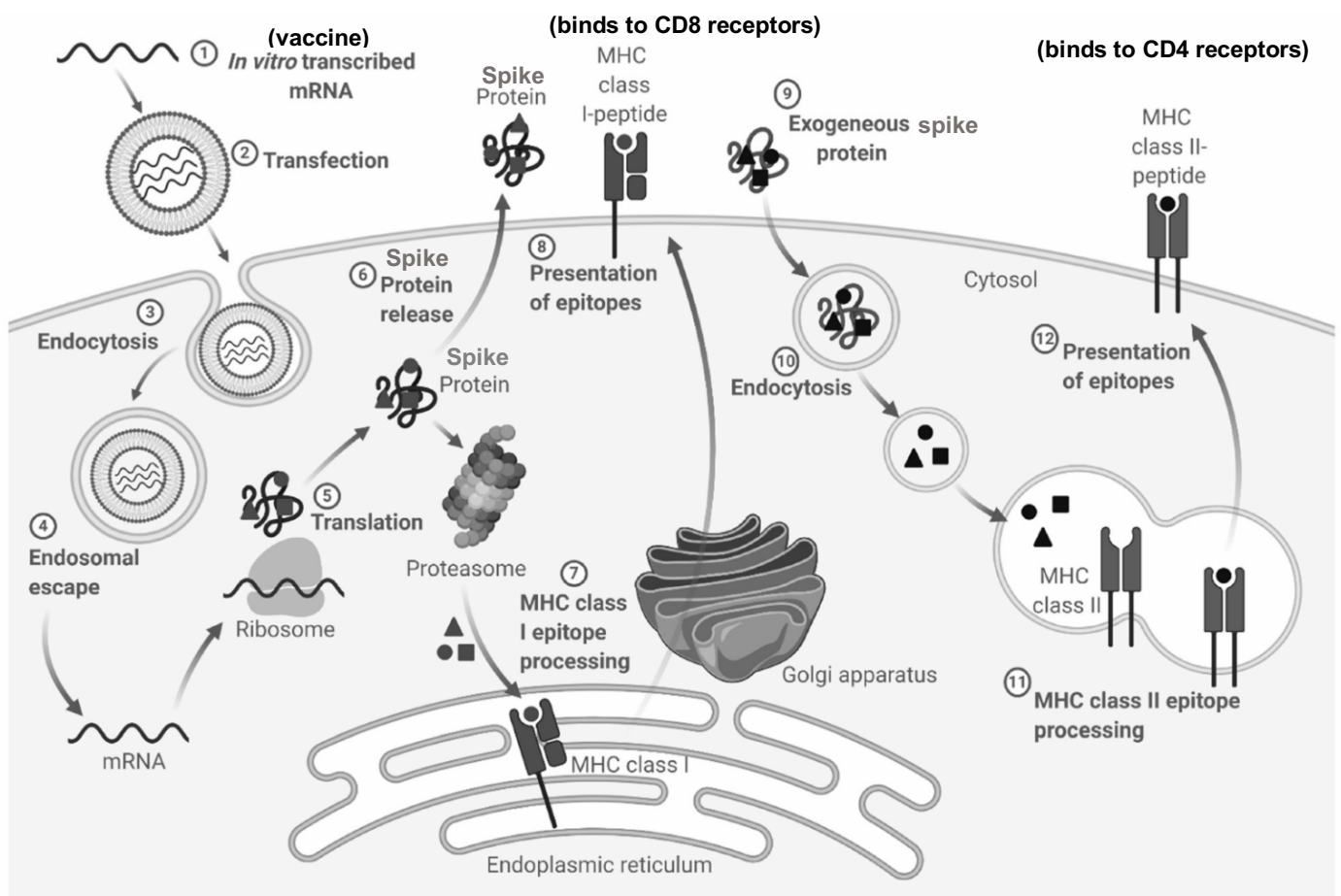


Fig. 3.1

- (i) With reference to the information above and Fig. 3.1, outline how an mRNA vaccine allows an acquired immune memory to be built against future coronavirus infection.

[L2]

[4m]

1. **PRESENTATION:**

From Fig 3.1, the cell that has taken up the mRNA vaccine translates it to produce the **spike proteins** which are digested into antigen fragments presented on **MHC class I**. Spike proteins are also released out of the cell.

2. **SELECTION:**

Some of the exogenous spike proteins are taken in and processed to be presented on MHC class II; **CD4+ T-cells** with the ***complementary antigen receptors would bind to these antigen fragments presented on MHC class II and become **activated into T helper cells**,

3. **ACTIVATION by T-helper:**
which would then **activate (BOTH) B and CD8+ T cells** with the same complementary antigen receptors, CD8+ T cells bind to antigen fragment on MHC class I to get fully activated
4. **MEMORY:**
T and B cells that are activated proliferate and some of these cells differentiate into memory cells that **can recognise the same spike proteins** present on coronavirus in future infections.
5. AVP; **B-cells also take up the exogenous spike proteins released out of the cell and present respective antigen fragment to T helper cell**, triggering its own activation.
6. AVP; **exogenous spike proteins** can be **taken up by macrophages / dendritic cells** to present them on MHC to **CD4+ cells with complementary antigen receptors** and activate them into T-helper cells

(ii) Explain why, despite not being considered as living organisms, viruses can evolve.

[L3]

[2m]

1. While viruses are not living organisms, they are made of **biomolecules: nucleic acids encapsulated by proteins** which are coded for by the same nucleic acids. As the **nucleic acids can undergo mutations, this generates the variation** of which selection pressure can act on (raw materials of evolution).
2. Different phenotypes of viruses **undergo selection pressure of hosts' immune systems**, resulting in specific strains that have "selective advantage" and are able to replicate more copies of itself.
3. Selected viruses are able to infect and replicate within host cells, **passing down the same genes / alleles to copies of viruses which are identical to itself**.
4. AVP; **OWTTE**

Concerns over the efficiency of vaccines was not just because of the potential ineffectiveness against new viral strains, but the consequences of expanding partial dosage of the vaccines in an attempt to increase the number of people able to receive the vaccine.

Concerns were raised about having a fraction of the population with partial immunity to the virus could potentially increase selection for vaccine-escape variants, ultimately undermining vaccine effectiveness.

(iii) Discuss if there should be concerns about the effectiveness of vaccines, given that the coronavirus can evolve.

[L3]

[3m]

1. Yes; because when the coronavirus evolves, the **current vaccines may no longer be effective** against the same virus as **the immune system will no longer recognise the mutated protein**.
2. Yes; because **partial immunity could potentially select for viruses that escape the selection** of the immunity provided for by vaccines, **producing viruses that are resistant to the immune memory** built by the vaccine.

3. No; **having a high amount of people vaccinated could create herd immunity** which decreases the spread of virus; this would **lower the spread of virus and potential variation required for evolution**.
4. No; partially vaccinated people would **still have a higher immunity against actual viral infection**; this would allow them to **reduce viral growth and hence potential genetic variation required for evolution**.

[Total: 13]

Section B

Answer **one** question in this section.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

- 4 (a) Describe the molecular structure and functions of enzymes, and explain why high fevers can be dangerous for the body. [13]
- (b) Explain the significance of membranes in ATP synthesis in living organisms, and discuss the significance of ATP synthesis in organisms that lack such membranes. [12]
- 5 (a) Outline how eukaryotic somatic cells normally divide, and describe how this process could develop into one that results in cancerous cells instead. [14]
- (b) Despite selection pressures selecting for certain phenotypes, some phenotypes continue to persist in a population.
- Explain how both genetic and environmental factors contribute to phenotypic expressions in eukaryotes and suggest how phenotypes which are **selected against** could be preserved in a population. [11]

4a [L1/2]

A Molecular structure of enzymes (5 max)

1. Enzymes are **large globular proteins** with a specific conformation; their primary structures **usually contain more than a hundred amino acids**
2. The **active site** of an enzyme comprise the 'contact' and 'catalytic' residues where **contact/binding residues** are responsible for the specificity of the enzyme and **form a shape that is complementary** to the shape of substrate.
AVP) Binding residues have R-groups that **allow the enzyme to bind to substrate temporarily** and hold the substrates in place for the reaction to occur.
3. **Catalytic residues** are responsible for the **ability of the enzyme to catalyse a particular chemical reaction**; acts on the chemical bonds in the substrate.
4. The remaining amino acids i.e. **structural residues**, which make up the bulk of an enzyme, **function to maintain the correct globular shape** of the enzyme molecule.
5. In the tertiary structure of an enzyme, **R-groups of different amino acids** which may be far apart in the primary sequence **form hydrogen, ionic, disulphide bonds and hydrophobic interactions** to maintain the overall specific **3D conformation** of the enzyme.
6. AVP; Enzymes **may** have a quaternary structure, when the overall 3D conformation requires two or more polypeptide chains formed by ionic, etc bonds for it to function
R: Descriptions that imply all enzymes have quaternary structures (factually incorrect)
7. AVP; Enzymes **may** also require a non-protein component, co-enzymes / co-factors to bind to the protein structure to complete its functional 3D conformation;
Some enzymes are not protein in nature, e.g. peptidyl transferase comprises entirely RNA.
8. AVP; Enzymes are usually soluble proteins, amino acid residues with hydrophobic R-groups are found within the interior of the protein while amino acid residues with hydrophilic R-groups are found on the exterior

B Functions of enzymes (4 max)

1. Enzymes **lowers the activation energy required for a specific biological reaction** by forming enzyme-substrate complexes when it collides successfully with substrate molecule(s).
2. **[Idea of bond-forming]**
Within the enzyme-substrate complex, the **enzyme molecule holds the substrate molecules in the precise orientation** that increases the likelihood of effective collision between the substrate molecules.
3. **[Idea of bond-breaking]**
The precise active site of the enzyme **may also cause certain bonds in the substrate molecule to be placed under stress**, increasing the likelihood for the bonds within substrate to be broken.
4. **[Idea of bond-forming]:**
The active site may change the charge on the substrate, alter the distribution of electrons within the bonds of the substrate or **cause other changes that will increase / change the reactivity of the substrate molecules**.
5. AVP; Ref to lock and key & induce fit hypothesis in description of how enzymes **bind** to substrate to **carry out a reaction** (function)
R: description of what is lock and key / induced fit models with no reference to function as that does not answer the question.

C Why fever is dangerous (4 max):

1. Enzymes are **responsible for many important biological reactions** within the body; when they cannot function, many such reactions cannot occur e.g. ATP synthesis in aerobic respiration.
2. Enzymes best **functions at an optimal temperature** where the rate of diffusion of enzyme and substrate molecules are high enough with minimal enzymes denaturing.
3. High fever means **excessive heat / high kinetic energy disrupts the intra-molecular bonds** such as hydrogen bonds, ionic bonds, and hydrophobic interactions, which are needed to **stabilise the secondary and tertiary structures** of the enzyme molecule.
4. The enzyme molecule **unfolds**, and the **precise shape of the active site is lost**, also known as **enzyme denaturation** and it is irreversible.

QWC: all parts of the question answered adequately with consistently clear use of scientific language

4b [L2/3]

Significance (max 8m)

A. Significance of membranes;

1. Membranes are formed from **phospholipid** molecules that form a **bilayer** with hydrophilic **phosphate heads facing outwards** towards aqueous medium while the **hydrophobic hydrocarbon tails** face inwards and form a **hydrophobic core**.
2. Structural outline
 - i. Chloroplasts contain **double membrane** with an **outer membrane**, and an **inner thylakoid membrane forming stacks** called grana.
 - ii. Contains **double membrane** with an **outer membrane** and an **inner membrane forming finger-like projections** called cristae
3. **High number of folds** for **increased surface area to volume ratio** for increased attachment of proteins for oxidative phosphorylation / photophosphorylation while low volume maintains high concentration of H^+
4. **Inner membrane** also contains **various proteins** for
 - i. **Photophosphorylation** in **chloroplasts**
 - ii. **oxidative phosphorylation** for aerobic respiration in **mitochondria**
5. **Proximity of the proteins** within the membrane **allows for higher efficiency** of enzymatic reactions and ETC for photophosphorylation / oxidative phosphorylation
6. Various **electron carrier proteins** that **form the electron transport chain** for transferring electron and **pumping of H^+** from one space to another
 - i. stroma into thylakoid space
 - ii. mitochondria matrix into intermembrane space
7. Hydrophobic cores of double membranes being **impermeable to H^+** allows for **compartmentalisation (space) for building up of high H^+ gradient**
8. The build-up of a proton gradient helps contribute to a **proton motive force** for **chemiosmosis of H^+** through the **ATP synthase** for ATP synthesis
9. AVP; **membranes contain light-harvesting proteins** i.e. chlorophyll **to capture light energy** for photophosphorylation in chloroplasts
10. AVP; membranes allow for compartmentalisation of enzymatic reactions e.g. Krebs cycle which produces the NADH and $FADH_2$ required for oxidative phosphorylation (link to O.P. is required)
11. AVP; membranes contain pyruvate transporters to allow for pyruvate from glycolysis to enter mitochondria

B. Anaerobic respiration / Fermentation (Max 5):

1. Organisms that lack such membranes can only carry out anaerobic respiration, where **ATP is synthesised via** processes such as **substrate level phosphorylation in glycolysis**

2. **NAD⁺ is required in glycolysis** and organisms that lack such membranes need to carry out metabolic processes that **allow for the (re)generation of NAD⁺**, such as alcoholic fermentation or lactic acid fermentation.
3. **As they lack electron transport chains**, the only way to regenerate NAD⁺ is **via fermentation** where **organic compounds are reduced to organic acids/alcohol**.
4. Substrate level phosphorylation is significant in such organisms **because it is the only process that produces ATP since they lack membranes** (e.g. obligate anaerobes)
5. This suggests that **such ATP synthesis have evolved before the evolution of membranes adapted for more efficient ATP synthesis** (i.e. oxidative and photo- phosphorylation)

QWC: all parts of the question answered adequately with consistently clear use of scientific language

5a [L1/2] (Novel)

A. Mitosis (max 6m)

1. Eukaryotic somatic cells normally divide by mitosis (nuclear division) and cytokinesis. Before mitosis begins, **DNA is replicated first** during **interphase**.
2. During prophase, **nuclear envelope disintegrates**, and **chromatin fibres become more tightly coiled**, **condensing into discrete chromosomes** and spindle fibres begin to form.
3. During metaphase, **kinetochore proteins bind to centromeres** and **spindle fibres connect to both kinetochore** and from pole to pole, **chromosomes are aligned singly at the metaphase plate**
4. During anaphase, the **centromere divides** and the **sister chromatids from each chromosome separate** and **move to opposite poles** of the spindle
5. During telophase, **chromosomes** arrive at each pole and **uncoil** to become back to chromatin fibres, and **nuclear envelope reforms**.
6. Cytokinesis then occurs, where either a cleavage forms for animal cells or a cell plate forms for plant cells, where **the cell membrane close around the two separate nuclei** and **separates the two daughter cells**.

B. Development of cancer (9 max):

1. **One named carcinogen** e.g. X-rays directly damage DNA or indirectly through free radicals which **causes mutations** in the DNA.
2. **As DNA is replicated**, **any (and many) mutations** that control the cell cycle and cell division are also replicated and **accumulate** over time, occurs in **a single cell line**;
3. Mutations include a **loss of function mutation** of **both alleles/copies of tumor suppressor genes** e.g. *RB* gene involved in **preventing the stimulating activity of cellular proto-oncogenes or oncogenes**; OR **insensitivity to growth signals resulting in uncontrolled cell proliferation**;
Reject: no mention of both alleles
4. **Loss of function mutation of both copies** (accept recessive) **of tumour suppressor gene** involved in **activating apoptosis resulting in the prevention of repairing of DNA damage**, cell apoptosis, arrest of cell cycle, e.g. *p53* hence **allowing accumulation of mutations**;
Accept: ECF for both alleles from pt 3
5. **Gain in function mutation** in one of **proto-oncogenes into oncogenes**, causing **over production of growth stimulating proteins** OR **self-sufficiency in growth signals** e.g. *ras* gene;
6. Gain in function mutation, or mutation causing the **activation of telomerase gene allows for the cell to replicate beyond the normal replicative potential**, OR **overcome the end replication problem in normal cell replication** i.e. Hayflick limit;
7. Gain in function mutation, or mutation causing the **activation of angiogenesis** and hence **increased vascularisation providing a continual supply of oxygen and other nutrients, sustaining the growth** of tumour / cancerous cells;
8. Loss of function mutation / mutations causing the **inactivation of genes involved in cell-to-cell adhesion / recognition**, allowing for **metastasis** i.e. cancer cells escape from tumor, circulate in the blood and lymph and **invade / spread to** other areas of body;

For points 6-8, mutation or at least activation / inactivation of genes need to be mentioned;

E.g. reject "angiogenesis occurs"

9. Accumulation of these mutations **allow the cancerous cells to replicate bypassing the G1 and G2 checkpoints** of cell cycle, resulting in **uncontrolled cell division**; (accept M if included)
10. **AVP**; related to the development of cancer

QWC: all parts of the question answered adequately with consistently clear use of scientific language

5b [L2/3]

A. Genetic factors (max 6m)

1. Genes that are **expressed** are **transcribed and translated into proteins** which **directly affects** or **phenotypes** as many of these proteins are involved in metabolic processes.
2. **Different forms / sequences** of the same genes, **alleles**, determine the type of protein synthesised and therefore **variation of phenotype expressed**.
3. Most eukaryotes are **diploid** organisms that have two sets of chromosomes each containing the same gene; hence every cell of a eukaryote contains **at least 2 sets of genes at each gene locus**.
4. In a heterozygote with one dominant and one recessive allele for the same gene, the **expression of the dominant allele masks** that of the **recessive allele**.
5. Some phenotypes can be **controlled by multiple gene loci**, such phenotypes are typically in a **continuous variation**.
6. Phenotypes that are **controlled by more than one gene can exhibit epistasis**, where the **expression of one gene locus is masked by the genotype of another gene locus**.
7. AVP; some genes could exhibit **co-dominance**, where **both dominant alleles are expressed equally** and neither masks the other, e.g. I^A and I^B alleles for blood type.
8. AVP; some genes could exhibit **incomplete dominance** where there is **some expression of the recessive alleles** despite the presence of the dominant allele; this would **result in a mixed phenotype** that is unequal.
9. AVP; phenotype can also be affected by **(a) control elements** and **(b) genes coding for transcription factors** that **change the phenotypic expression of genes** when they are mutated.

B. Environmental factors: (max 3m)

1. Environment can also contribute to phenotypic expressions by **affecting the proteins within organisms** such as through temperature, food, chemicals (**at least 1 example demonstrated, max 2**);
 - a) e.g. bee larvae and diet
 - b) e.g. Himalayan rabbits
2. Hence both environment and genetic factors contribute to phenotypic expressions via the proteins that determine the phenotype, and **environmental factors can potentially mask the expression of genetic factors** e.g. genetically identical twins with different weight / skin tan
3. Phenotypes that are controlled by multiple genes loci are usually **affected by environmental factors more** as each protein can be influenced as compared to those controlled by a single gene locus.

C. Preservation of phenotypes selected against (max 7m)

1. *Idea of* masked genes / alleles **not being subjected to selection pressure** and hence not removed from gene pool
2. These **alleles are still being passed down to offspring** and **may be expressed again** in the next generation

Heterozygote protection

3. **Phenotypes coded for by recessive alleles** that are selected against can persist in **heterozygotes** as they are completely masked by the dominant allele and will not be subjected to selection pressure.

Masking by epistasis

4. Genes that are being selected against by the selection pressure could **exist in phenotypes that have it being masked by other genes** e.g. dominant epistasis

Masking by mutations in non-coding sequences

5. **Mutations could occur in regulatory sequences** (mainly **enhancers**) which stops the expression of dominant genes.
6. These genes that are being passed down could then be **expressed again upon re-activation of the regulatory genes / other mutations that allows expression.**

Masking by environment factors

7. Some **alleles could be masked by environmental factors** such as **food / medication**
 - a. e.g. deficiency in certain essential nutrients in the body that can be substituted by diet / medication
 - b. e.g. insulin deficiency being substituted by insulin injection that allows one to not be affected by diabetes

QWC: all parts of the question answered adequately with consistently clear use of scientific language